

Experiment 2:

Aim: To demonstrate 2 class confusion matrix using python

Program:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.datasets import load_wine
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score, f1_score, precision_score, recall_score,
classification_report, confusion_matrix

wine = load_wine()
data = pd.DataFrame(data=wine.data, columns=wine.feature_names)
data['Target'] = wine.target
data = data[data['Target'] != 2] # Filter for 2 classes

x = data.drop('Target', axis=1)
y = data['Target']
x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.3, random_state=1)

model = DecisionTreeClassifier(random_state=1)
model.fit(x_train, y_train)
y_pred = model.predict(x_test)

accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)

class_report = classification_report(y_test, y_pred, target_names=wine.target_names[:2])
print("Classification Report:\n", class_report)

precision = precision_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)
print(f"Precision: {precision:.2f}")
print(f"Recall: {recall:.2f}")
print(f"F1 Score: {f1:.2f}")

conf_matrix = confusion_matrix(y_test, y_pred)

plt.figure(figsize=(8,6))
sns.heatmap(conf_matrix, annot=True, fmt='d', cmap='PuBuGn', xticklabels=wine.target_names[:2],
yticklabels=wine.target_names[:2])
plt.xlabel("Predicted Label")
plt.ylabel("True Label")
plt.title("Confusion Matrix")
plt.show()
```

```
Accuracy: 0.9743589743589743
*** Classification Report:
      precision    recall  f1-score   support

   class_0       1.00      0.94      0.97        17
   class_1       0.96      1.00      0.98        22

   accuracy                0.97        39
  macro avg       0.98      0.97      0.97        39
 weighted avg       0.98      0.97      0.97        39
```

Precision: 0.96
Recall: 1.00
F1 Score: 0.98

